MIND OVER MATTER

Reviewing the Six Pillars of Effective Teaching and Learning at EHS: A Brush-up Guide

For our students to master and retain knowledge not just for an individual assessment, but for their lives, they need more than just information. Our students need a learning environment that is intentionally designed to support the way the human brain works.

The six strategies we're highlighting - Managing Cognitive Load, Retrieval Practice, Spacing, Interleaving, Feedback, and Metacognition — are the research-backed **pillars** of just such an environment. When we embed these into our teaching, we're not just covering content; we're building the foundation for lifelong learning.

As we mentioned in August, these are techniques that almost everyone is using in some ways already, and the more intentional we can be about implementing them, the more likely our students are to learn — and retain - what we want them to take away from our courses.

Practice: Think of these pillars not as separate rules, but as an integrated system that can be applied at different stages of your planning.

As you plan out your course or unit, you can use ***spacing*** *and* ***interleaving*** *to map out the key concepts to which you'll circle back.* ***Retrieval practice*** *can be regularly sprinkled in, ideally on a daily basis, to solidify learning by getting it out of students' brain.*

In the classroom, you're actively managing ***cognitive load*** *by breaking down complex topics into digestible chunks. (This is, like, our whole job!) You're fostering* ***metacognition*** *by asking students to reflect on their own thinking and learning processes.*

Post-assessment, you provide ***feedback*** *that helps students understand their strengths and areas of growth. This feedback then (****metacognitvely****!) informs next steps, including which concepts need more* ***retrieval****, leading to a new cycle of learning.*

01

Cognitive Load Challenge:
In a world that always asks for more, consider how you can do one thing...less! Simplify a complex concept by breaking it down into steps, allow students to explore a visual without you yapping at the same time, or simplify homework so it focuses on the primary skills you want to foster, without any extra.

02

Retrieval Practice Challenge:
Start your next class with a one-minute "brain dump." Ask students to write down everything they can remember from their previous lesson. Don't grade it-just let them practice retrieving the information.

03

Spacing Challenge:
Identify a key concept taught three weeks ago. In your next lesson, design a low stakes activity (poll, activity, or prompt) that forces students to revisit and integrate that older concept into their current skill/topic/unit.

04

Metacognition Challenge:
Replace standard content exit tickets with metacognitive prompts that encourage students to reflect on their learning process, such as how their thinking about the topic changed or what strategy they chose and why.

05

Feedback Challenge:
On a assignment, commit to giving only one "star" (a specific thing they did well) and one "wish" (a specific point for improvement). Keep it concise and focused on a single actionable next step. (Do less to do more!)

06

Interleaving Challenge:
For the next homework, Intentionally mix up the types of problems or questions so they don't appear in the order you taught them. This causes students to differentiate between problem-solving methods, ensuring that they know the purpose of each tactic they use.

Meet with Luke or Jamie

If you have any queries or questions,
please do not hesitate to contact Luke or Jamie

Share a win